

2.1 Analytical Question: The Impact of Data Scaling

Problem statement (unchanged).

Discuss the sensitivity of Principal Component Analysis to the scaling of the input features. Explain why performing PCA on a raw, unstandardized dataset can lead to misleading principal components. Use a hypothetical example (e.g., one feature measured in kilometers and another in millimeters) to illustrate your point. Conclude by outlining the conditions under which you would choose not to standardize the data before applying PCA.

Solution (reformatted):

Principal Component Analysis is fundamentally a variance-maximization technique. Its primary objective is to find orthogonal directions (principal components) that capture the maximum amount of variance in the data. This direct dependence on variance makes PCA highly sensitive to the scale of the input features.

Explanation of sensitivity

The principal components are the eigenvectors of the data's covariance matrix $\mathbf{S}$. The diagonal entries S_{ii} represent the variance of the i -th feature. If features are measured on vastly different scales, the feature with the largest numerical variance will dominate $\mathbf{S}$ and, consequently, the PCA solution. The first principal component will be almost entirely aligned with the axis of that high-variance feature — not because the feature is necessarily the most important structurally, but because its units yield much larger numerical variance.

Hypothetical example

Consider two features:

- Feature A (body length), measured in kilometers, typical value 0.02 km, variance $\approx 10^{-5} \text{ km}^2$.
- Feature B (fin width), measured in millimetres, typical value 2000 mm, variance $\approx 10^5 \text{ mm}^2$.

If PCA is performed on the raw data, $\mathbf{S}$ will have diagonals roughly 10^{-5} and 10^5 . The total variance is almost entirely from Feature B, so the first principal component will lie nearly along the fin-width axis. Changing units (e.g., converting body length to mm) will change which feature dominates — showing that PCA on unstandardized data is unit-dependent and can produce arbitrary, misleading results.

Role of standardization

Standardization (z-scoring each feature: subtract mean, divide by standard deviation) transforms features to zero mean and unit variance. PCA on standardized data effectively works on the correlation matrix rather than the covariance matrix, making the result invariant to original feature scales. Standardization is a modeling choice that encodes the prior belief that features are a priori equally important.

When not to standardize

You might choose **not** to standardize when:

- **Commensurable features:** All features share the same units/scale and direct comparison of variances is meaningful (e.g., pixel intensities).

- **Meaningful zero & scale:** The absolute scale or zero-point has physical meaning (e.g., Kelvin temperatures in some contexts).
- **Sparsity considerations:** For very sparse, count-based data (e.g., raw word counts), standardization can destroy sparsity; alternatives like TF-IDF or other scalings are preferred.

In short: standardize by default unless domain knowledge justifies preserving raw scale.

2.2 Numerical Problem: Eigendecomposition and Data Projection

Given covariance matrix

$$\mathbf{C} = \begin{pmatrix} 13 & -4 \\ -4 & 7 \end{pmatrix},$$

(a centered dataset).

(a) Eigenvalues

Solve $\det(\mathbf{C} - \lambda \mathbf{I}) = 0$:

$$\det \begin{pmatrix} 13 - \lambda & -4 \\ -4 & 7 - \lambda \end{pmatrix} = (13 - \lambda)(7 - \lambda) - (-4)(-4) = 0.$$

Expanding:

$$\lambda^2 - 20\lambda + 75 = 0 \quad \Rightarrow \quad (\lambda - 15)(\lambda - 5) = 0.$$

Hence

$$\lambda_1 = 15, \quad \lambda_2 = 5.$$

(b) Orthonormal eigenvectors

For $\lambda_1 = 15$:

$$(\mathbf{C} - 15\mathbf{I})\mathbf{v} = \begin{pmatrix} -2 & -4 \\ -4 & -8 \end{pmatrix} \mathbf{v} = \mathbf{0}$$

gives $-2v_1 - 4v_2 = 0 \Rightarrow v_1 = -2v_2$. Choose $v_2 = 1 \Rightarrow$ unnormalized eigenvector $(-2, 1)^\top$.

Normalizing:

$$\mathbf{v}_1 = \frac{1}{\sqrt{(-2)^2 + 1^2}} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 5$:

$$(\mathbf{C} - 5\mathbf{I})\mathbf{v} = \begin{pmatrix} 8 & -4 \\ -4 & 2 \end{pmatrix} \mathbf{v} = \mathbf{0}$$

gives $8v_1 - 4v_2 = 0 \Rightarrow v_2 = 2v_1$. Choose $v_1 = 1 \Rightarrow$ unnormalized eigenvector $(1, 2)^\top$. Normalizing:

$$\mathbf{v}_2 = \frac{1}{\sqrt{1^2 + 2^2}} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

(As expected for a symmetric matrix, $\mathbf{v}_1^\top \mathbf{v}_2 = 0$. Note eigenvectors may be sign-flipped without changing span.)

(c) Proportion of variance explained

Total variance $= \lambda_1 + \lambda_2 = 15 + 5 = 20$.

Proportions:

$$\text{Prop}_{\text{PC1}} = \frac{15}{20} = 0.75 \quad (75\%), \quad \text{Prop}_{\text{PC2}} = \frac{5}{20} = 0.25 \quad (25\%).$$

(d) Transforming a new centered point

Given $\mathbf{x} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, form the eigenvector matrix with columns in order of descending eigenvalue:

$$\mathbf{V} = \begin{pmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix} \quad (\text{columns: } \mathbf{v}_1, \mathbf{v}_2).$$

Transformed coordinates (principal-component coordinates) are

$$\mathbf{x}' = \mathbf{V}^\top \mathbf{x}.$$

Compute each component:

$$\begin{aligned} x'_1 &= \mathbf{v}_1^\top \mathbf{x} = \frac{-2}{\sqrt{5}} \cdot 2 + \frac{1}{\sqrt{5}} \cdot 3 = \frac{-4+3}{\sqrt{5}} = -\frac{1}{\sqrt{5}}, \\ x'_2 &= \mathbf{v}_2^\top \mathbf{x} = \frac{1}{\sqrt{5}} \cdot 2 + \frac{2}{\sqrt{5}} \cdot 3 = \frac{2+6}{\sqrt{5}} = \frac{8}{\sqrt{5}}. \end{aligned}$$

So

$$\mathbf{x}' = \begin{pmatrix} -\frac{1}{\sqrt{5}} \\ \frac{8}{\sqrt{5}} \end{pmatrix} \approx \begin{pmatrix} -0.4472 \\ 3.5777 \end{pmatrix}.$$